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Climate Change Impacts and Future Risk on UK Seahorse Species, Short-Snouted *Hippocampus hippocampus* and Long-Snouted *Hippocampus guttulatus*

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ABSTRACT

For the two European seahorse species, the short-snouted *Hippocampus hippocampus* and the long-snouted *Hippocampus guttulatus*, there are knowledge gaps that need further research. These research gaps become increasingly pressing under climate change, where uncertainty in how coastal ecosystems will change is compounded by uncertainty in how these seahorses will respond to changing pressures. Under climate change, these species could experience northward range shifts, expanding their range in United Kingdom (UK) coastal waters, potentially requiring the UK to take a leadership role in European seahorse conservation in the future. This review aims to synthesise current scientific research to provide an overview of how these seahorse species are likely to respond to climate change. Using the most recent Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6), predictions for future change along with the likelihood and severity of seahorse species response are combined to provide a confidence ranking in the climate change risk for these species in the UK. When considering individual factors, such as sea surface temperature, these seahorse species show some resilience, but climate change is the cumulative impact of multiple stressors, which existing research has not been able to capture. Overall, further research on seahorse response to environmental variables is needed across Europe.

1 | Introduction

The coastal waters of the United Kingdom (UK) have two native seahorse species, the short-snouted seahorse *Hippocampus hippocampus* and the long-snouted seahorse *Hippocampus guttulatus*. Seahorses are members of the Syngnathidae family. Both seahorse species are found along the coasts of North Africa, throughout the Mediterranean, and around to the coasts of Portugal, Spain and northern France, extending to northern Germany (Figure 1). The distributions of both species in the UK are concentrated along the southern English coast, from Cornwall to Kent, with some sightings identifying possible populations along the Welsh coast, the Thames Estuary and near Bangor, Northern Ireland. In Europe, *H. guttulatus* is more commonly sited and is the focus of most European studies, but in the UK (as shown in Figure 1), *H. hippocampus* is described as more commonly sighted, with a greater spatial area of sightings (Bluemel et al. 2023; Pierri et al. 2022). Though in some regions these two species can co-occur in up to 97% of the same area (Bosso et al. 2024), it cannot be assumed that their biology, ecology or chemical tolerances, for example, are identical.

Seahorses are often described as having sparse distributions and low population densities (Correia 2022; Gomes et al. 2023; Gristina et al. 2021; Pierri et al. 2022). They also have a limited swimming ability, meaning they use their prehensile tail to grip

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FIGURE 1 | Geographic distribution of short-snouted *Hippocampus hippocampus* and long-snouted *Hippocampus guttulatus* seahorses based on observational data and research locations. *Source*: Pierri et al. (2022).

onto holdfasts to withstand currents (Correia et al. 2013). Their limited swimming ability may contribute to another feature of these species: high site fidelity (Garrick-Maidment et al. 2010; Woodall et al. 2017). Globally, most seahorse species are commonly associated with complex habitats such as seagrasses, corals or mangrove forests (Choi et al. 2012; Pinault et al. 2018; Scapin et al. 2018). *H. guttulatus* is commonly associated with seagrass beds (Bluemel et al. 2023; Garrick-Maidment et al. 2010; Woodall et al. 2017) but has also been associated with other holdfast habitats such as tunicates (Correia 2022) and artificial hard substrates with abundant algae (Gristina et al. 2021). The association with more structurally complex habitats is linked to holdfast, shelter from predators and camouflage opportunities, as well as increased abundance of prey and shelter from which to ambush prey (Kitsos et al. 2008; Woodall et al. 2017). Although *H. hippocampus* can also be found near or within complex habitats, there have been sightings of *H. hippocampus* in more sparse habitats (Correia et al. 2018; Gristina et al. 2022) as well as with artificial holdfasts (Correia 2022; Correia et al. 2018). Though these sightings cannot be determined as a preference of *H. hippocampus* for more open habitats, these instances again emphasise that the biological and ecological requirements cannot be assumed the same for both species, requiring the need for individual assessment.

Despite potential differences between these two species, the pressures each species are facing are presumed similar. Both species are listed under the Convention on International Trade in Endangered Species of Wild Flora and Fauna (CITES) Appendix II due to targeted fishing for alternative medicine, curios or illegal aquaria stocking (Koning and Hoeksema 2021). Both species are currently, as of June 2024, classed as 'data deficient' on the International Union for the Conservation of Nature (IUCN) Red Lists of Threatened Species (Pollom 2017; Woodall 2017), though for the Mediterranean, both species are classified as 'near threatened' (Pollom 2016a, 2016b). The lack of data regarding distributions and population dynamics means that there are gaps in the understanding of population trends in the UK. Under climate change, it is currently unknown how the direct (e.g., temperature increase, ocean acidification and sea level rise) and indirect (e.g., increased sedimentation or freshening of coastal areas due to potential increased winter and summer rainfall) pressures will affect UK seahorse populations.

Here, this review aims to synthesise existing research on climate change impacts on UK seahorse species, while also providing a confidence ranking of the climate change risk for these species (Table 1). The climate change pressures reviewed here were chosen based on existing European-wide reviews (Hodapp et al. 2023; Monteiro et al. 2023; Pierri et al. 2022) and research undertaken by the UK's Natural England and the Centre for Environment, Fisheries, and Aquaculture Science (CEFAS) (Bluemel et al. 2023). Using the Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6), predictions for regional changes in European coastal systems (Fox-Kemper et al. 2021; Lee et al. 2021; Ranasinghe et al. 2021) will be compared to research on seahorse response to these changing pressures. Where possible, this will also be supported by UK-based studies of climate change predictions to ensure more localised models are following overall regional predictions and thus represent the most probable impact for UK seahorse species. The confidence ranking given for each pressure in this review is determined from the certainty or confidence provided by the IPCC's own predictions in combination with the extent of evidence for and the severity of seahorse response.

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TABLE 1 | Confidence rankings for climate change risk to UK seahorse species justification.

Confidence rank	Classification justification
Low	Low certainty in future change of discussed pressure, no or little evidence for seahorse response to changing pressure
Medium	Medium or high certainty in future change of discussed pressure **WITH** no or little evidence for seahorse response to changing pressure **OR**; Low or medium certainty in future change of discussed pressure **WITH** extensive evidence for seahorse response to changing pressure
High	High certainty for future change of discussed pressure, extensive evidence of seahorse response to changing pressure

2 | Current Pressures

Due to a lack of available data, it is hard to determine how climate change is currently affecting UK seahorse populations. From climate modelling, it is clear that global mean sea surface temperature (SST) has changed by 0.88 [0.68 to 1.01]°C from 1850–1900 to 2011–2020 and 0.60 [0.44 to 0.74]°C from 1980 to 2020 (Fox-Kemper et al. 2021). Marine heatwaves (MHWs) are defined as a period of five or more days where temperatures are warmer than the 90th percentile based on a 30-year historical baseline period (Hobday et al. 2016). MHWs have likely doubled in frequency between 1982 and 2016 (Fox-Kemper et al. 2021). Changes in salinity have more localised patterns, with some areas becoming more saline and some experiencing freshening (Fox-Kemper et al. 2021; Wells et al. 2022). Observational data and modelling have also shown an increase in relative sea level around Europe from 1900 to 2018, the regional mean change being 1.08 mm year^{-1} (Ranasinghe et al. 2021). In a global context, there is clear evidence that oceans are acidifying (Caldeira and Wickett 2003; Fox-Kemper et al. 2021). There is some evidence from the North Sea (30 years of data) and UK coastal sites (6 years of data) that pH is decreasing, but trends are spatially and temporally variable (Williamson, Turley, and Ostle 2017). In the current observed data, due to low data quality, there is very low confidence that storms have increased (Wolf, Woolf, and Bricheno 2020).

Research suggests that nonclimate-related pressures, such as pollution, non-targeted fishing (Lawson, Foster, and Vincent 2017) and seagrass habitat decline, are affecting seahorse species globally, which are assumed to be operating in the UK currently also, but more research is needed regarding this. For example, in the UK, it is estimated that at least 44% of UK's seagrasses have been lost since 1936, 39% since the 1980s (Green et al. 2021), which represents a loss of potential habitat for seahorses (Bluemel et al. 2023). It is unclear whether other biotic and abiotic factors such as temperature or water quality would have allowed for more abundant seahorse populations from 1930 to 1980, however. There is little data tracking historical pollution events across coastal ecosystems in the UK, but chemical (Delunardo et al. 2020; Domínguez-López et al. 2022; Prévot D'Alvise, Ascensio, and Richard 2024), light (James and Heck 1994) and noise pollution (Anderson et al. 2011; Palma et al. 2019) are current known anthropogenic pressures for seahorses elsewhere, so it can be assumed that seahorses are also facing these pressures in UK coastal waters.

3 | Future Pressures

The elusive nature of seahorses, their low population densities and the research challenges for conducting experimental studies, such as the transport and storage of individuals without the death of specimens, makes it difficult to understand short- and long-term impacts, as well as the influences of multiple stressors on these species. Climate change predictions for coastal regions also vary in confidence, meaning it is uncertain how climate change will influence the physical and chemical characteristics of coastal areas. Moreover, in coastal areas, predicting the impacts of global mechanisms can be difficult due to the high influence of local factors at the coast (Fanjul et al. 2019; Wells et al. 2022). To increase confidence in our understanding of what might happen in the future in UK coastal waters and how seahorse populations may respond, existing research must be applied. An overview of *H. hippocampus* and *H. guttulatus* responses to predicted potential climate change impacts is listed in Table 2. Here, this section will focus on the impact of individual pressures on the two UK seahorse species, to assess the state of science.

4 | Direct Pressures

4.1 | Sea Surface Temperature

For European coastal waters, average annual SST is predicted to increase between 1°C and 3°C by 2100, with 1°C reflective of representative concentration pathway (RCP) 4.5, and the 3°C warming of RCP8.5 (Ranasinghe et al. 2021). The prediction has high confidence, so UK coastal waters are high likely to experience warming by 2100. A UK-based study predicted that an increase of 3°C is expected for the English Channel even under a medium emissions scenario (Mahaffey et al. 2023).

Based on laboratory studies from European populations, including those from Portugal (Aurélio et al. 2013; Costa et al. 2023; Faleiro et al. 2015), Spain (Planas, Quintas, and Chamorro 2013) and Italy (Bosso et al. 2024), it is predicted that even under extreme warming (+3°C), seahorse species in the UK will persist. This is because the UK populations are inhabiting areas that have the minimal reproductive conditions (i.e., temperature requirements for spawning are only met for a small portion of the year) due to these populations currently being towards the northernmost limit of their range. For example, a laboratory study of Spanish *H. hippocampus* and *H. guttulatus* found that a 15°C–18°C temperature increase led to significantly lower egg clutch size in both *H. hippocampus* and *H. guttulatus*, when compared to 15°C–21°C, and in *H. guttulatus* at 15°C, the yolk volume within eggs was significantly reduced (Planas, Quintas,

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TABLE 2 | Positive and negative impacts of climate change for native UK seahorse species with a confidence ranking based on the certainty of the future prediction as well as the extent of the evidence base for impact on the seahorse species. *#Hh* and *#Hg* refers, respectively, to number of studies from *H. hippocampus* and *H. guttulatus* from which positive and negative impacts have been evidenced. #O refers to number of studies from other seahorse species from which positive and negative impacts have been obtained. Confidence ranking is determined based on impacts and studies presented here as well as the literature discussed in the main text. Individual studies are listed in the Supplementary Information file.

Impact	Positive impact	Negative impact	*#Hh*	*#Hg*	#O	Confidence
SST increase	• Increased feeding activity • Increased reproductive behaviour • Higher quality eggs produced	• Unsure of survival or stress impacts for increased warming over longer periods	3	6	0	High
Ocean acidification		• Lethargy behaviour displayed in feeding, socialising and reproduction	0	1	0	Medium
Sea level rise	• Could migrate to other shallow water habitats, minimal impact	• Increased depths could increase wave action at the coasts displacing seahorses • Could effect seagrass distributions, in turn effecting seahorses	0	0	0	Medium
Storminess increases		• Juveniles intolerant of higher flows could be displaced from home populations	2	3	0	Medium
Storminess decreases	• Reduced juvenile mortality or displacement could improve reproduction rates		1	0	0	Low
Salinity increases	• Potential salinity increases not likely to induce stress behaviour or death	• Stress behaviour induced in high salinity areas	0	0	1	High
Salinity decreases	• Potential salinity decrease not likely to induce stress behaviour or death	• Freshening can influence zooplankton communities, a key food source for seahorses	0	0	3	High
Dissolved oxygen decreases		• Low oxygen and hypoxia immediately triggers lethargic behaviour reducing feeding, socialising, and reproductive behaviour	2	0	0	Medium
Nutrient enrichment		• Olfactory cues are important for reproduction, which pollution could interfere with • Eutrophication events have caused population declines of up to 99% over 5 years in Spain	0	1	2	Low
Sedimentation increases		• Reduced light can interfere with feeding ability • Seahorses rely on visual cues for reproduction	4	2	1	Medium
Seagrass distribution increases	• Increased food sources • Increased availability of holdfast, refuge, and camouflage habitat		1	2	0	Low
Seagrass distribution decreases	• Not explicitly linked to seagrasses only, they can use other habitats	• Loss of holdfast, refuge and camouflage habitat • Decreased food sources	2	5	0	Medium
Food resource changes	• If climate change causes more plankton blooms, more food sources will be available for juveniles • Seahorse diet seems adaptable	• Any change to plankton blooms could be out of sync with seahorse breeding season resulting in decreased food availability	2	2	0	Low

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and Chamorro 2013). The breeding season for these seahorse species can be from March to October under the right phototactic and thermal conditions, but currently, temperatures in southern England and Wales only begin to exceed 16°C from July to September (CEFAS temperature modelling, H. Tidbury, pers. comms., 2024). Under climate change, there is potential for thermal conditions suitable for seahorse reproduction to expand further northward (AquaMaps 2019a, 2019b; Hodapp et al. 2023; Monteiro et al. 2023). Increased temperature has also been shown to increase food uptake for adult *H. guttulatus*, up to 30°C (Faleiro et al. 2015). Similar increases in feeding, and potentially breeding activity, were observed in a study of *H. guttulatus* up to 24°C (Costa et al. 2023).

Not all results from temperature-focused studies are conclusively positive. A study of *H. hippocampus* found that there was no change in activity (i.e., feeding, swimming, resting and holdfast activity) under different temperatures, though this experiment only tested up to 25°C (Gristina et al. 2022). Though *H. guttulatus* has been recorded unstressed up to 30°C, this had been for short periods of time (i.e., < 4 weeks), and so there is concern for survival at prolonged higher temperatures, as during the experiment, increased feeding did not directly correspond with increased growth in weight (Costa et al. 2023). The metabolism of juvenile *H. guttulatus* was found to decrease at temperatures beyond 28°C, suggesting that increased water temperatures can have variable effects on each life stage (Aurélio et al. 2013). In a thermal study looking at adult *H. guttulatus* response to warming intervals: 18°C–26°C, 18°C–28°C and 18°C–30°C, the increases in oxygen consumption and metabolism were smallest in the 18°C–30°C warming response study, suggesting that a maximum thermal tolerance threshold was being approached after 30°C (Aurélio et al. 2013).

Based on two studies from Portugal, *H. guttulatus* demonstrated the ability to withstand +12°C of warming from the average summer temperature (18°C) of the Sado Estuary to the tested 30°C laboratory conditions (Aurélio et al. 2013; Faleiro et al. 2015), whereas for *H. hippocampus*, of the published research available, the upper thermal limit tested is 25°C (Gristina et al. 2022), which are also from a Portuguese population. Thus, it could be estimated, based on the average summer temperature (18°C), *H. hippocampus* could potentially withstand warming of +7°C.

These studies, however, are from more southern populations where marine heatwaves can currently reach 24°C and are predicted to exceed 30°C by the end of the century (Aurélio et al. 2013; Faleiro et al. 2015). In the UK, average summer temperature throughout the existing range of *H. hippocampus* and *H. guttulatus* is 16.73 ± 0.86°C (based on NOAA satellite data from www.seatemp.org). Based on this current average summer temperature, theoretically, *H. guttulatus* could withstand an additional 12°C warming, reaching 28.73°C, whereas *H. hippocampus* could withstand average summer warming to 23.73°C. Under current climate change predictions for coastal warming by 2100, summer temperature for southern English and Welsh coastal waters will warm to 19.73°C on average under a high concentration pathway (i.e., RCP8.5, +3°C). Thus, future thermal stress for UK seahorse species is unlikely, due to their exhibited temperature range tolerance and, second, because future warming predicted for this northern area in their range is still within the thermal range that they are currently experiencing elsewhere. The warming predicted for coastal SST in the UK will likely increase thermal habitat suitability and may lead to a range expansion if other factors are also suitable (Hodapp et al. 2023; Monteiro et al. 2023). Overall, the assessment of SST impact on UK seahorse species has high confidence based on the extensive research and the high confidence in the future climate change predictions.

4.1.1 | Marine Heatwaves

Concurrently with increases in SSTs, there is potential for MHWs to change as well. Recent models suggest that for Northern Europe, annual MHW frequency will increase by 1–2 MHW per decade, where the current average is one or two events per year (Ranasinghe et al. 2021). Average length of MHWs are between 10–15 days, but it is less certain how duration and intensity will change under climate change (Ranasinghe et al. 2021). The predictions are also not spatially even. For example, intensity per event is expected to increase in the North Atlantic and the Mediterranean but decrease in the Atlantic off the British Isles (Ranasinghe et al. 2021). Other studies have suggested that as the English Channel already experiences frequent MHWs, it is likely to experience longer, more intense MHWs in the future (Simon et al. 2023). The increasing frequency of MHWs, as well as potential for an increase in duration and intensity, could be detrimental for seahorses in the UK. MHWs have the ability to interfere with the timing of phytoplankton blooms (Simon et al. 2023), which are linked to distributions of zooplankton, with some taxa such as copepods being important to juvenile and adult seahorse diet (Kitsos et al. 2008; Randazzo et al. 2018). If southern England experiences more frequent, higher intensity and longer heatwaves, where temperatures reach above 28°C, there is potential for this to affect the metabolism of juvenile seahorses (Aurélio et al. 2013) and may influence feeding behaviour of adults (Costa et al. 2023; Faleiro et al. 2015). Under extreme warming, where SST may increase by +3°C, resulting in average summer SST of 19.73°C, MHWs would have to reach extreme temperatures of over 30°C in order to negatively influence the feeding and reproductive behaviour of *H. guttulatus*, and increase by ~6°C to influence *H. hippocampus*, based on current published knowledge. Although there is high confidence in the IPCC's predictions of more frequent MHWs, how UK seahorse species will be directly and indirectly affected has less confidence due to the lack of certainty surrounding the intensity, duration and timing of future MHWs.

4.2 | Ocean Acidification

Ocean acidification is driven by decreases in pH from the global average of ~8.07 ± 0.02 pH (Jiang et al. 2019) to become more acidic, caused by changes in seawater chemistry from carbon absorption. Seawater pH is expected to decrease, and this falling trend will begin to emerge by 2050 under RCP8.5 (Ranasinghe et al. 2021). This prediction from the IPCC has medium to high confidence, and smaller studies have found similar predictions. For example, pH is expected to drop anywhere between 8.05 and 7.6 by 2100 (Garrard and Beaumont 2014). One study of adult

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H. guttulatus manipulated both temperature and pH (−0.5 pH) to better study the impact of potential climate change, where seahorses were exposed for 1 month to temperatures of 18°C, 26°C and 30°C, whereas pH was maintained at current levels and another set of these temperatures were combined with a −0.5 pH change, creating six experiments in total (Faleiro et al. 2015). Under all temperatures where there was a pH decrease, all seahorses decreased in feeding and ventilation activity, resulting in lethargy. Ultimately, the study concludes that although adult *H. guttulatus* may be able to withstand potential future SST changes, the combination with increasing ocean acidification could affect their reproduction and survival (Faleiro et al. 2015). There are no studies on pH tolerance of other syngnathid species, but in other studies of marine fish species, pH has been shown not to inhibit reproduction or survival (Lopes, Faria, and Dupont 2020), though the effects of ocean acidification are highly variable across the diverse range of marine fish species (Esbaugh 2018). Without more research, especially for *H. hippocampus*, it is impossible to conclude with high confidence the impact potential future changes in ocean acidification could have on UK seahorses.

4.3 | Sea Level Rise

Regional sea level rise for Europe is estimated to be 0.4–0.8 m by 2100, though locally, there are likely large deviations in actual sea level change (Ranasinghe et al. 2021). In the future, there is also likely to be an increase in the frequency of larger coastal flooding events. For example, by 2050, a 1 in 100 year event will occur approximately every 5 years (Ranasinghe et al. 2021). The increase in depth is presumably of little impact on seahorses as despite their sedentary behaviour, they are mobile, and there is some speculation that some populations may have seasonal migrations to deeper waters, potentially to avoid winter storms (Garrick-Maidment et al. 2010). There is potential that sea level rise will increase wave energy, which will impact local coastal ecosystems. For *H. guttulatus*, who are more strongly associated with seagrasses (Bluemel et al. 2023; Garrick-Maidment et al. 2010; Woodall et al. 2017), sea level rise could influence seagrass distribution as increased depths could reduce light penetration, a key factor in seagrass distribution (Tang and Hadibarata 2022). Overall, sea level rise will have a low direct impact on UK seahorse species, but there could be potential consequences through increased wave energy and loss of associated habitats (see Section 5.3), though the certainty of these impacts is very limited, resulting in a medium overall ranking (Table 2).

4.4 | Storminess

There are no IPCC predictions for storminess under future climate change, which is due to the challenges of modelling storms currently. More local research, concentrated on the UK, has some predictions, but with variable results. Some models show that North Atlantic storms over the 21st century will reduce in frequency, as well as reduce wave heights on average, but there may be an increase in the more extreme wave heights (Wolf, Woolf, and Bricheno 2020). Others, assessing predicted changes in weather patterns under climate change, have shown that under RCP8.5, more days with impactful winter storm conditions may occur, but there is far greater variability within the occurrence of these weather patterns (Pope et al. 2022). Similarly, another study of UK weather patterns showed that potential future changes will not be spatially or temporally even (Perks et al. 2023). They identified that the southern English coast seems most vulnerable to increases in the frequency of weather patterns associated with winter storminess, particularly under high emission pathways (i.e., RCP8.5) (Perks et al. 2023). Based on these studies, although there is a lack of confidence in the predictions, a potential increase in storminess under climate change is possible.

If storminess increased, there would be variable impacts to coastal habitats due to increases in current, wave energy and wave height, which will influence UK seahorse species. A general current tolerance for both species has been determined as 0.25–0.8 m s^{-1} (H. Tidbury pers. comms., 2020). Adult *H. guttulatus* has been shown to withstand currents of 1 m s^{-1}; the juveniles, however, have a lower tolerance to withstand current increases as they showed increased grasping behaviour in currents as low as 0.1 m s^{-1} (Correia et al. 2013). Though, in the Ria Formosa Lagoon, Portugal, where both *H. hippocamus* and *H. guttulatus* are found, maximum current speed can reach 1.25 m s^{-1} (Pacheco et al. 2010). For adult *H. hippocampus*, some studies have emphasised that any change in wave action may be detrimental to the species, as they could be removed from their habitat, or their ideal substrata for tail anchoring could be removed or lost (Nash, Sabatini, and Ballerstedt 2021). Overall, the uncertainty surrounding whether storminess will increase and to what magnitude, and the resilience of UK seahorse species is too great to confidently assess this particular climate change risk.

4.5 | Salinity

Future predictions of salinity changes for coastal waters are variable. For the UK, studies suggest that surface salinities, at the coast and more open marine waters, are freshening (Dye et al. 2020, 2013). The English Channel, for example, has been predicted to have an annual mean change of −0.08 ± 0.25 PSU by the end of the century (i.e., 2069–2098) relative to 1960–1989 averages (Dye et al. 2020). This is distinctly less than other areas, such as the Northern North Sea, which is estimated to experience −0.62 ± 0.65 PSU annual mean change in surface salinity by the end of the century. All studies emphasise, however, that there is considerable uncertainty regarding future salinity changes.

Currently, there are no direct studies of the impact of salinity changes on *H. guttulatus* and *H. hippocampus*. It is possible, however, to assess the salinities across the species' distributions to understand levels of tolerance. For example, in the Marano and Grado Lagoon of the Northern Adriatic Sea, salinity ranges between 5 and 40 PSU, where *H. hippocampus* and *H. guttulatus* are known to inhabit (Bettoso et al. 2013). Moreover, some salinity studies exist for other seahorse species, which could infer the direct reaction of UK seahorse species to salinity changes. A study of common seahorse *H. kuda* juveniles found that the upper tolerance was 50 ppt and growth and survival of juveniles was reduced between 15 and 20 ppt

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(Hilomen-Garcia, Delos Reyes, and Garcia 2003). Beyond both of these boundaries, death was observed. Another study of slender seahorse *H. reidi* found that for juveniles, growth (i.e., weight and height) decreased with increasing salinities between 10 and 35 PSU (da Hora et al. 2016). A separate study of juvenile *H. erectus* was shown to display stress behaviour at low salinities (15 ppt) (Huang et al. 2020). Interestingly, the Kynsna seahorse *H. capensis* has been reported to tolerate salinities of up to 59 ppt for 7 days (unpublished data cited in Whitfield 1995).

Despite evidenced impact at salinity extremes, the scale of change expected in the UK will likely not exceed the extremes at which seahorses experience stress or death (Dye et al. 2013). UK seahorses currently inhabit areas of varying salinities. For example, seahorses have been surveyed in the Solent, where salinities can reach 35 PSU (Townend 2008) and in the Thames Estuary where salinities can be between 10 and 20 PSU (Ferrero, Debenham, and Lambshead 2008). Overall, the confidence ranking for the impact of salinity changes on seahorses is high, due to UK coastal waters predicted to experience minimal changes in salinity as well as seahorses being found in a variety of saline habitats currently in the UK.

5 | Indirect Pressures

5.1 | Deoxygenation

Average annual mean dissolved oxygen concentration in the global ocean is predicted to decline between 1.5% and 4% by 2090 under all RCPs (Mahaffey et al. 2023). For UK shelf waters, models predict that annual mean oxygen concentration will decline most strongly in the North Sea regions and in the Celtic sea, estimating between 5.6% and 5.9% decrease by 2100 under RCP8.5, which could see minimum monthly mean dissolved oxygen concentrations declining from ~7.75 mg L^{-1} pre-2000s levels to < 6 mg L^{-1} by 2100 (Mahaffey et al. 2023). Large areas of the Celtic and Irish Seas, as well as the English Channel, are predicted to be at risk of oxygen deficiency in the future (Mahaffey et al. 2023). Extreme deoxygenation, hypoxia, has not been detected in UK marine waters but has occurred in northwest European shelf seas, including the North Sea and Baltic Sea (Mahaffey et al. 2023). Although hypoxia has not been detected, periods of oxygen deficiency have been detected in UK waters of the North Sea and Celtic Sea, but the impact of low oxygen on the marine ecosystem is relatively understudied (Mahaffey et al. 2023).

One study found that *H. hippocampus* can withstand acute hypoxia (~27% dissolved oxygen) for approximately 7 h (Gomes et al. 2023). The 7 h did not necessarily represent a lethal limit for the seahorses but was just the boundary of the experiment. The impact of acute hypoxia on the seahorses was reduced activity, including a significant reduction in metabolism and ventilation rates, as well as reduced feeding and socialising behaviour (Gomes et al. 2023). The study concludes that more long-term consequences to increased hypoxia events or exposure are unknown but their short-term lethargic response makes them vulnerable to starvation or predation, reducing their chances of survival (Gomes et al. 2023). Nash, Sabatini, and Ballerstedt (2021) similarly commented that for *H. hippocampus*, a poor swimmer with a small home range, hypoxia is likely detrimental to the species. Though, some seahorses do have the ability to migrate (up to 150 m per day for *H. guttulatus*; Caldwell and Vincent 2013) and could possibly move from hypoxic areas. Overall, the predictions for a reduction in dissolved oxygen in UK coastal waters are confident, but the spatial disparities and strength of the reductions are more uncertain. The lack of evidence of the impact of deoxygenation on seahorse species reduces the confidence of this assessment.

5.2 | Increased Runoff

5.2.1 | Nutrient Enrichment

Under climate change, winter precipitation is expected to increase, as well as some more intense summer rainfall, which will increase surface runoff throughout river catchments, potentially leading to increased nutrient loading in coastal areas (Hannaford 2015; Watts et al. 2015). There is potential for nutrient loading to directly and indirectly impact UK seahorse species, but the extent to which they will be impacted is challenging to predict. For example, in the Mar Menor coastal lagoon, Spain, where there is a population of *H. guttulatus*, eutrophication from nutrient loading has been attributed as a cause of population decline, resulting in a 99% population reduction over 5 years (Vivas et al. 2024). It is possible that nutrient enrichment via pollution inputs could interfere with reproductive behaviour, as in pot-bellied seahorse *H. abdominalis*, it has been shown that females use olfactory cues to select mates during the reproductive season (Bahr et al. 2012) and olfactory cues are also important for *H. erectus* (Lin et al. 2021). Indirectly, nutrient enrichment has been found to negatively affect seagrass beds if the nutrient loading facilitates eutrophication due to a reduction in dissolved oxygen and light penetration (Pascal et al. 2022). Due to its association with seagrass beds, there is potential for *H. guttulatus* to have reduced access to key habitats if nutrient enrichment worsens in the future. Nutrient loading also influences phytoplankton blooms, which can influence zooplankton distribution, potentially influencing the ability of juvenile seahorses to feed. Overall, whether nutrient enrichment will increase is uncertain due to a link with land management practices, knowing the chemical composition of nutrient loading events as well as their frequency is uncertain, and how seahorses will be impacted is also uncertain, making the confidence in this climate change risk assessment low.

5.2.2 | Sedimentation

Similarly to increased nutrient enrichment from increased runoff, sedimentation can also increase (Hughes et al. 2022). For both UK seahorse species, research has shown the importance of light for reproduction, which can be reduced by sediment loading (Planas, Quintas, and Chamorro 2013). For example, for both *H. hippocampus* and *H. guttulatus*, light was shown to be a strong controlling factor in triggering reproduction and extending the photoperiod artificially, extended the breeding season of both species allowing reproduction to happen earlier, even at low temperatures (Planas, Quintas, and Chamorro 2013).

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In *H. hippocampus*, visual cues are well documented as being crucial to mate selection, courtship and mate bonding, all behaviours vital to reproduction, and so increased sedimentation is thought to be detrimental to *H. hippocampus* reproduction (Nash, Sabatini, and Ballerstedt 2021). In *H. erectus*, light was found to be a key factor in prey capture success as it decreased with increasing turbidity (James and Heck 1994). Based on this, it is suggested that for *H. hippocampus*, increased turbidity would reduce feeding and growth, influencing sexual maturity and reproduction (Curtis et al. 2017). Despite these potential impacts, *H. hippocampus* is currently found in estuarine areas with high turbidity in the UK, such as the Thames Estuary (Bluemel et al. 2023). Although this may suggest a turbidity tolerance for *H. hippocampus* to these conditions, there is a lack of data to clarify whether *H. hippocampus* inhabits the area for the entirety of its lifecycle (i.e., reproduction, feeding, juvenile hatching and first feeding) or whether they have just been observed there. Based on a lack of evidence for understanding the potential extent of sedimentation increases under climate change, in combination with some extensive evidence for light being important to seahorses, there is the potential for sedimentation to adversely affect UK seahorse species.

5.3 | Habitat Distribution Changes

It is widely recognised globally that habitat destruction, loss or fragmentation is one of the main threats for most seahorse species (Pinault et al. 2018; Scapin et al. 2018). The most common cause of habitat loss, destruction or fragmentation is direct anthropogenic impact, such as development, boating or trawling (Green et al. 2021; Harasti 2016). For example, in Australia, some populations of *H. whitei* have declined by up to 97% between 2009 and 2015 (Harasti 2016). In combination with these direct anthropogenic impacts, climate change is altering the physiochemical composition of the habitats seahorses occupy.

Similar to other seahorse species, *H. guttulatus* and *H. hippocampus* are associated with seagrass and complex habitats (Bluemel et al. 2023), whereas *H. hippocampus* can also be found across more open habitats (Bosso et al. 2024). Seagrass habitats are facing direct impacts under climate change, through SST, salinity, pH and dissolved oxygen changes, as well as changes in sea level rise and storminess, which ultimately will affect *H. guttulatus* populations through habitat association. In combination with sea-level rise, which can reduce light penetration and negatively impact seagrasses (Tang and Hadibarata 2022), by 2100, northern Europe could experience median shoreline retreat of around 50 m under RCP4.5 and up to 80 m under RCP8.5 (Ranasinghe et al. 2021). Depending on local geology (i.e., bays or headlands), there is a possibility that seagrass habitats could shift inland and thereby be minimally affected by sea-level rise. In some areas, however, inland migration may not be possible, and seagrass habitats could be lost. Although there is some evidence that some climate change impacts, such as ocean acidification, could benefit seagrass growth (Garrard and Beaumont 2014), other impacts, such as increased nutrient loading, have variable impacts across the microecosystem of seagrass habitats. Seagrasses can buffer these impacts to an extent, but it is clear that there is a limit to buffering capacity of seagrasses when it comes to nutrient loading and eutrophication (Pascal et al. 2022).

Along the southern English coast, where current UK seahorse populations are concentrated, there is potential for seagrass distributions to decline as SST is expected to increase, MHWs become more frequent, and potentially, storminess increases. If seagrass distributions decline, then this could be detrimental for *H. guttulatus*. Juvenile *H. guttulatus* has strongly exhibited feeding behaviour while attached to holdfasts (Gristina et al. 2021), demonstrating the importance of these habitats for both shelter and feeding. For seahorse populations across Europe, the association with seagrass beds has been attributed to the diverse macroinvertebrate community, key food sources, that seagrass beds provide (Woodall et al. 2017). Any declines in seagrass bed distribution also represent a loss of food sources. A study of *H. guttulatus* from Mar Piccolo, Italy, found, from eDNA analysis of faecal matter, that adults fed on invasive zooplankton as well as native, reflecting their adaptability to changing environmental conditions as invasive species migrate into a system (Lazic et al. 2023). Potentially, the food sources provided by seagrasses are not strictly critical to survival of *H. guttulatus*, though juveniles of both short- and long-snouted seahorses are less adaptable when it comes to dietary requirements (Blanco, Quintas, and Planas 2011; Palma et al. 2023). Declines in seagrass beds would also reduce holdfast availability and perhaps make seahorse species more vulnerable to predation or storminess (Correia et al. 2013; Nash, Sabatini, and Ballerstedt 2021). Though, if seagrass bed distribution increased under climate change, it could be beneficial for seahorse species, due to the increased availability of holdfasts, food sources and shelter from predation and storminess. Despite an association with these habitats, there is potential that seahorses will be less affected by changes in seagrass beds, but more information is needed, resulting in a medium confidence assessment.

5.4 | Food Source Distribution Changes

As well as affecting habitat, climate change may influence food source availability for both juvenile and adult seahorse species, due to changes in coastal water chemistry. Previous studies have made an association between *Hippocampus spp.* distributions and chlorophyll-a concentrations (used as an indicator of phytoplankton food source availability), with some identifying that concentrations above 10 mgC/m^3 represent optimal habitat (Bluemel et al. 2023). There are very few UK-based studies assessing how climate change will influence phytoplankton and zooplankton communities. Some suggest that declines in salinity due to freshwater inputs from increased runoff have resulted in zooplankton declines already in some areas (Wells et al. 2022). Others suggest that by 2100, primary production in the English Channel will increase, and consequently, zooplankton biomass will also increase (Chust et al. 2014). Though studies in other areas have found that plankton bloom increases are not uniform for every species under climate change and would depend on individual species tolerance to changing environmental factors (e.g., SST increase and salinity decreases) (Hedberg et al. 2024). Other broader studies suggest that under IPCC global warming scenarios, copepods will experience poleward shifts, with earlier seasonal peaks, and changes in spatial patterns of biodiversity (Villarino et al. 2015). A long-term study of plankton around the UK coasts has shown that SST has increased abundance of meroplankton but decreases in other groups such as holoplankton

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(dominated by small copepods) (Bedford et al. 2020). Decreases in copepods were also found in Holland et al. (2023). Under climate change, it is hard to confidently assess how food sources will change for seahorses. Existing literature suggests that adult seahorses are adaptable in their diets (Lazic et al. 2023), though juveniles are less adaptable. Seahorses are also expected to have a poleward shift in distribution under climate change (AquaMaps 2019a, 2019b; Hodapp et al. 2023; Monteiro et al. 2023), and perhaps, this movement will synchronise with distribution changes in phytoplankton and zooplankton blooms (Villarino et al. 2015). Overall, due to the challenges of modelling changes in plankton communities and an uncertainty surrounding the adaptability of UK seahorse species, there is low confidence currently in the understanding of how food source changes will affect *H. hippocampus* and *H. guttulatus*.

6 | Remaining Challenges

For most potential climate change impacts, there is low confidence in the risk these impacts pose to UK seahorse species. This is mostly due to a lack of confidence in future predictions or lack of research and understanding on how UK seahorse species may respond to the pressure.

Monitoring, data collection and analysis of UK seahorse populations are perhaps the key current challenge. Without more consistent data collection and evidence sharing, a comprehensive understanding of UK seahorse species cannot be developed. This is particularly crucial as under climate change, there is potential for the UK coastal waters to become more suitable for seahorse species, as they potentially shift northwards and the south of their range in north Africa and the Mediterranean become uninhabitable (Monteiro et al. 2023). Without more comprehensive data on UK populations, it is hard to design and implement conservation action. From this review, however, one policy recommendation is to reduce sediment runoff in agriculturally-dominated river catchments that deliver sediments to the coast, due to extensive research demonstrating the importance of visual cues for seahorse reproduction.

Additionally, another key issue relating to these studies of northward range shifts is that the methodologies cannot accurately capture the complex, interacting multiple stressors occurring in marine and coastal environments. The approaches of Monteiro et al. (2023), for example, can be simplistic, using only a handful of climate change variables. The models do not include coinciding anthropogenic impacts, such as sedimentation increases or nutrient loading, as well as increased storminess, the latter being incredibly challenging to model. Without addressing the multiple stressors and localised pressures affecting local seahorse populations, it is likely that populations will not improve independently. For some seahorse populations, up to 90% of recruitment is found within 500 m of the original population (Andrade et al. 2016), reflecting the importance of addressing the local stressors for local populations. Another recommended potential conservation approach that could have far-reaching, holistic benefits across coastal ecosystems is to focus on the recovery and restoration of seagrass beds. For UK seahorses, this restoration could provide holdfasts, shelter and food sources (Pierri et al. 2022), as well as other benefits such as carbon sequestration (Hughes et al. 2022), but must be combined with ameliorating local stressors.

It is important to highlight that a key area that was not addressed during this review was the impact of invasive species on native seahorse species, the arrival of which are increasing under climate change (Mahanes and Sorte 2019). Across multiple ecosystems, invasive species can be detrimental to native flora and fauna (Elton, Simberloff, and Ricciardi 2020; Nunes et al. 2015), and that is inclusive of seahorses. For example, the arrival and spread of nonnative *Caulerpa prolifera* algae in the Formosa Lagoon, Portugal, have been suggested as a potential cause in population decline of *H. guttulatus* and *H. hippocampus* (Correia 2022). Invasive species could have direct impacts on seahorses, that is, through predation, or indirectly through the manipulation of the physiochemical composition of the surrounding habitats utilised by the seahorse species. A detailed review of the localised impacts of invasive species on seahorse species throughout Europe is thus needed.

7 | Conclusions

Current known pressures that directly impact both *H. hippocampus* and *H. guttulatus* seahorses are pollution, targeted and nontargeted fishing, as well as habitat degradation and fragmentation (Curtis et al. 2007; Delunardo et al. 2020; di Franco et al. 2020; Huang et al. 2020; James and Heck 1994; Lawson, Foster, and Vincent 2017). Under climate change, however, there is greater uncertainty in how these species will respond. Current research has focused on SST, showing that these species can tolerate temperatures up to 30°C for short periods. In a UK context, this suggests that SSTs will become more suitable for seahorses, perhaps leading to a northward range expansion. There is more limited research on the impacts of ocean acidification, deoxygenation and salinity on these species, evidencing the need for more research. More challenging yet is estimating the potential response to the cumulative impact of multiple climate change stressors, which is the more realistic nature of climate change that coastal ecosystems, as well as seahorse species, are facing. Ultimately, it is the lack of research on these species that has resulted in a low confidence ranking for many climate change pressures reviewed here. The knowledge gaps for these species are not only preventing an accurate assessment of climate change risk in the UK but the development of conservation action and planning.

Author Contributions

Reagan H. Pearce: investigation (lead), visualisation (lead), writing–original draft preparation (lead), writing–review and editing (equal). **Hazel Selley**: supervision (lead), writing–review and editing (equal).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors confirm that all data analysed during this study are included in this published article.

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